



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

MA1008-1 – CALCULUS AND LAPLACE TRANSFORMS

(For BT, CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. For the curve $\frac{r}{a} = e^\theta$, derivative of arc $\frac{ds}{d\theta} =$
A) $2r$ B) $\sqrt{2} r$
C) r^2 D) $\frac{r}{2}$
2. The function $f(x) = x^2$ satisfy Rolle's theorem in which of the following interval?
A) $[1, 2]$ B) $[0, 1]$
C) $[-1, 1]$ D) $[-1, 0]$
3. If ϕ be the angle between the tangent and radius vector at any point on the curve, $r = f(\theta)$ then $\cot \phi$ is equals to:
A) $r \frac{dr}{d\theta}$ B) $\frac{1}{r} \frac{dr}{d\theta}$
C) $\frac{d\theta}{r \frac{dr}{d\theta}}$ D) $\frac{1}{r} \frac{d\theta}{dr}$
4. The derivative of arc S for the curve $y = g(x)$ is $\frac{ds}{dy} =$ _____
A) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ B) $\sqrt{1 + \frac{d^2y}{dx^2}}$
C) $\sqrt{1 + \frac{dx}{dy}}$ D) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$
5. If $f(x)$ is a differentiable function and $f(1) = f(4)$ then
A) $f'(c) = 0$ for some $c \in (1, 4)$ B) $f'(c) = 0$ for some $c \in (1, 2)$
C) $f'(c) = 0$ for some $c \in (3, 4)$ D) $f'(c) = 0$ for some $c \in (2, 4)$
6. Given $u = x + y, v = xy$ then the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is :
A) $x + y$ B) 0
C) $x - y$ D) 1
7. If $z = 6x^2y^2 + 8x^2$ then $\frac{\partial z}{\partial x}$ is :
A) $6y^2 + 16x$ B) $6x^2 + 8x^2$
C) $12xy^2 + 16x$ D) $6x^2 + 6y^2 + 8x^2$
8. If $A = f_{xx}(a, b), B = f_{xy}(a, b), C = f_{yy}(a, b)$ then $f(x, y)$ will have a maximum at (a, b) if
A) $f_x = 0, f_y = 0, AC < B^2$ and $A < 0$ B) $f_x = 0, f_y = 0, AC = B^2$ and $A > 0$
C) $f_x = 0, f_y = 0, AC > B^2$ and $A > 0$ D) $f_x = 0, f_y = 0, AC > B^2$ and $A < 0$

9. If $L\{f(t)\} = \bar{f}(s)$, then $L\{\sin 2t\} = \text{-----}$
 A) $\frac{2}{s^2 - 2^2}$ B) $\frac{2}{s^2 + 2^2}$
 C) $\frac{s}{s^2 - 2^2}$ D) $\frac{s}{s^2 + 2^2}$
10. If $L\{f(t)\} = \bar{f}(s)$, then $L\left\{\frac{f(t)}{t}\right\} = \text{-----}$
 A) $\int_s^\infty \bar{f}(s) ds$ B) $\int_0^\infty \bar{f}(s) ds$
 C) $\frac{1}{s} \bar{f}(s)$ D) $\bar{f}(s)$
11. Laplace transform of $e^{-2t} \cos 5t = \text{-----}$
 A) $\frac{5}{(s+2)^2 + 5^2}$ B) $\frac{(s+2)}{(s+2)^2 + 5^2}$
 C) $\frac{5}{(s-2)^2 + 5^2}$ D) $\frac{2}{(s-3)^2 - 2^2}$
12. $L\{u(t-4)\} = \text{-----}$
 A) $\frac{e^{-4s}}{s^2}$ B) e^{4s}
 C) $\frac{e^{4s}}{s}$ D) $\frac{e^{-4s}}{s}$
13. Inverse Laplace transform of $(s+2)^{-2}$ is -----
 A) te^{-2t} B) te^{2t}
 C) e^{2t} D) e^{-2t}
14. Inverse Laplace Transform of $\log\left(\frac{s+3}{s+4}\right)$ is
 A) $\frac{1}{t}[e^{-3t} - e^{-4t}]$ B) $-\frac{1}{t}[e^{-3t} - e^{-4t}]$
 C) $-\frac{1}{t}[e^{-3t} + e^{-4t}]$ D) $[e^{-3t} - e^{-4t}]$
15. $L^{-1}\{e^{-as}\bar{f}(s)\} = \text{-----}$
 A) $f(t)u(t-a)$ B) $f(t-a)u(t)$
 C) $f(t-a)u(t-a)$ D) $f(t+a)u(t+a)$
16. If $f'(t)$ is continuous and $L\{f(t)\} = \bar{f}(s)$ then $L\{f'(t)\} = \text{-----}$
 A) $\bar{f}(s) - sf(0)$ B) $s\bar{f}(s) - f(0)$
 C) $s^2\bar{f}(s) - sf(0) - f'(0)$ D) $\bar{f}(s) - sf(0) - f'(0)$
17. $\sin(5x + 6y) \frac{\partial^3 z}{\partial y^3} + \frac{\partial^2 z}{\partial x^2} + xy = 0$ is a partial differential equation of degree -----
 A) 2 B) 1
 C) 0 D) 3
18. One dimensional wave equation is
 A) $u_t = c^2 u_{xx}$ B) $u_{tt} = c^2 u_x$
 C) $u_t = c^2 u_x$ D) $u_{tt} = c^2 u_{xx}$
19. Solution of the partial differential equation $\frac{\partial z}{\partial x} + \cos(3x - 2y) = 0$ by direct integration is
 A) $z + \frac{\sin(3x - 2y)}{3} = xf_1(y)$ B) $z - \frac{\sin(3x - 2y)}{3} = f_1(y)$
 C) $z + \frac{\sin(3x - 2y)}{3} = f_1(y)$ D) $z - \frac{\sin(3x - 2y)}{3} = f_1(x)$
20. The partial differential equation obtained from $z = f(x+y)$ by elimination of arbitrary function is
 A) $\frac{x^2 \partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ B) $\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$
 C) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$ D) $\frac{x^2 \partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = 0$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	5	L*2	2	1
b)	Using Lagrange's mean value theorem determine c for the function $f(x) = \sin^{-1} x$ in $[0, 1]$.	5	L2	1	1
c)	Using Maclurin's series expansion expand the function upto third degree terms $f(x, y) = \cos x \cos y$.	6	L2	2	1
2. a)	With the usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	5	L2	1	1
b)	Find the radius of curvature at any point on the curve $r = a(1 - \cos \theta)$.	5	L2	1	1
c)	If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$ the show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$.	6	L1	2	1
3. a)	State and prove Cauchy's mean value theorem.	5	L1	1	1
b)	Prove that the two curve $r = a\theta$ and $r = \frac{a}{\theta}$ intersects orthogonally.	5	L1	1	1
c)	If $u = \log(x^2 + y^2)$ prove that $u_{xy} = u_{yx}$.	6	L2	2	1
Unit – II					
4. a)	Find $L[e^{-3t} \sin 5t \cdot \sin 3t]$.	5	L1	3	1
b)	If $L(f(t)) = F(s)$, prove that $L(t^n f(t)) = \frac{(-1)^n d^n}{ds^n} (F(s))$.	5	L2	3	1
c)	Find $L^{-1} \left[\frac{s^2 + s - 2}{s(s+3)(s-2)} \right]$.	6	L1	4	1
5. a)	Find Inverse Laplace transforms of $\cot^{-1} \left(\frac{s}{2} \right)$.	5	L1	4	1
b)	Find the Laplace Transforms of $\int_0^t \frac{\cos at - \cos bt}{t} dt$.	5	L1	3	1
c)	If $f(t)$ is a periodic function with period T such that $f(t+T) = f(t)$ for all values of t, prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$.	6	L2	3	1
6. a)	Using the convolution theorem find $L^{-1} \left[\frac{1}{(s^2+1)(s+1)} \right]$.	5	L1	4	1
b)	Solve the differential equation by Laplace transform $x''(t) - 4x'(t) + 4x(t) = 4e^{-2t}, x(0) = -1, x'(0) = 4$.	5	L2	4	1
c)	Rewrite the following function using unit step function and find its L.T $f(t) = \begin{cases} \sin t, & 0 < t \leq \frac{\pi}{2} \\ \cos t, & t > \frac{\pi}{2} \end{cases}$	6	L2	3	1
Unit – III					
7. a)	Form a Partial differential equation for $z = (x-a)^2 + (y-b)^2 + 1$ by eliminating the constants.	5	L1	5	1
b)	Solve the following higher order partial differential equation by direct integration method $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$.	5	L2	5	1
c)	Derive one dimensional wave equation in the form $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$	6	L2	5	1
8. a)	Form partial differential equations by eliminating the arbitrary function from the equation $f(x^2 + 2yz, y^2 + 2zx) = 0$.	5	L1	5	1
b)	Solve the non linear partial differential equation $p^2 + q^2 = x + y$.	5	L2	5	1
c)	Solve the following linear partial differential equations by Lagrange's method $xp + yq = z$.	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
